CHRIS XU

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EDUCATION

University of California-San Diego (UC San Diego) September 2021 - June 2027
4rd year PhD student in Mathematics *La Jolla, CA*

- Interests: number theory, arithmetic geometry, Langlands program, homotopy theory
- Student-ran seminars I ran or attended (references in parentheses): p-adic modular forms (Gouvea), class field theory (Cassels-Frohlich), Weil II (Kiehl-Weissauer), abelian varieties (Bhatt), Falting's theorem (Bhatt-Snowden), the Fargues-Fontaine curve (Scholze), prismatic F-gauges (Bhatt), Falting's theorem (Lawrence-Venkatesh)
- Conferences attended: p-adic L-functions and Eigenvarieties (Notre Dame, 07/22), Arithmetic and Topology over Global Fields (UW-Madison, 10/22), Spring School on Eigenvarieties (Heidelberg, 03/23), Spring School in Arithmetic Statistics (CIRM, 05/23), LuCaNT (ICERM, 07/23), Workshop on p-adic Arithmetic Geometry (IAS, 11/23), Arizona Winter School 2024 (UofA, 03/24)
- Doctoral advisors: Kiran Kedlaya, Aaron Pollack

Massachusetts Institute of Technology (MIT) August 2017 - January 2021
B.S. in Mathematics *Cambridge, MA*

- Relevant coursework: linear algebra, differential equations, analysis, abstract algebra, algebraic number theory, algebraic geometry, algebraic topology, algebraic groups, elliptic curves, probabilistic combinatorics

CURRENT PROJECTS

Model-free quadratic Chabauty for modular curves

- Almost all current methods for finding rational points on modular curves involve first finding a plane model before implementing quadratic Chabauty. With Isabel Rendell, I am currently developing a way to work directly with the moduli of elliptic curves without using any defining equations.

Special cycles on exceptional groups

- With Aaron Pollack, I am investigating relations between cohomology classes of certain special cycles on G_2 (resp. E_8), which involve producing half-integral weight classical modular forms (resp. quaternionic modular forms on G_2) via theta lifting.

Construction of supersingular mod p representations of GL_2 of a p-adic field

- With people from Arizona Winter School 2025, we are using known instances of Langlands reciprocity to construct supersingular representations with a prescribed Serre weight.

SELECTED PAPERS

Rigorous expansions of modular forms at CM points, I: Denominators January 2025

- Describes an algorithm to compute the power series expansion of a weight 2 cusp form at a CM point, with rigorous determination of the coefficients.

Skelet #17 and the determination of BB(5) February 2024

- Proves non-halting of a certain Turing machine dubbed "Skelet #17", arguably the most difficult remaining obstruction to proving Scott Aaronson's conjecture that BB(5) = 47,176,870.

UNDERGRAD

Here are three mini-projects I worked on during my undergraduate.

MIT Department of Mathematics June 2020 - August 2020
Undergraduate Researcher, UROP+ Program *Cambridge, MA*

- Computed cohomologies of determinant twists of a certain ring spectrum.
- Learned significant amounts of homotopy theory on the side, working with graduate mentor Robert Burklund.

MIT Department of Mathematics June 2019 - August 2019
Summer Program in Undergraduate Research (SPUR) *Cambridge, MA*

- Computed the traces of CM values of nonholomorphic weight 0 Maass forms.
- Built off of previous work of Bruinier and Funke for traces of modular functions.
- Collaborated with graduate mentor Yongyi Chen every day over a six week period.

MIT Department of Mathematics June 2018 - August 2018
Undergraduate Researcher, UROP+ Program *Cambridge, MA*

- Studied the signatures of irreducible matrix representations.
- Conjectured a polynomial pattern using *Atlas of Lie Groups and Representations*.
- Assisted in its proof.

AWARDS

$25,000 UW-Madison Van Vleck Scholarship Winner 2012

Math Olympiad Summer Program (MOSP) Qualifier 2015

Putnam Honorable Mention 2018, 2019